

Josh Miao

Renaming operator is the goat

1. # reservations / boat

 $\sigma_{bid, bname, COUNT(bid)} \rightarrow num_reservations(boats \bowtie reserves)$

2. Sailors who reserved All Red boats

 $RedBoats \leftarrow \pi_{bid} (\sigma_{color='red'}(boats))$ $Sailor_boat\ Pairs \leftarrow \pi_{sid, bid}(Reserves)$ $\pi_{sname, sid} ((Sailor_boat\ Pairs \div RedBoats) \bowtie Sailors)$ 3. Sailors who ONLY reserved red boats (≥ 1 reservation) $Sailors_reserved \leftarrow \pi_{sid}(Reserves)$ $Sailors_nonred_res \leftarrow \pi_{sid}(Reserves \bowtie \sigma_{color \neq 'red'}(Boats))$ $\pi_{sid} ((Sailors_reserved - Sailors_nonred_res) \bowtie Sailors)$

5. Sailors who've never reserved red boat

 $Sailor_red \leftarrow \pi_{sid}(Reserves \bowtie \sigma_{color='red'}(Boats))$ $\pi_{sid, sname} ((\pi_{sid}(Sailors) - Sailor_red) \bowtie Sailors)$

6. Average age of Sailors w/ rating 10

$\sigma_{AVG(age) \rightarrow average_age(\sigma_{rating=10}(Sailors))}$